COMP 3721 Practice Midterm

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Student Name:	
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Set:	
Exam Duration:	
Total Marks:	

Important Notice Regarding This Practice Midterm

1. This practice midterm is provided solely for review purpose.
2. The questions included in this document are for practice only and may or may not appear on the actual midterm examination.
3. This practice set is not intended to be comprehensive. It does not guarantee coverage of all topics, concepts, or question types that may be assessed in the midterm examination.
4. Students are responsible for studying all course materials, including but not limited to lecture slides, assignments, quizzes, discussions, and any other resources provided during the term.
5. Students are encouraged to use this practice midterm as a supplementary review tool and to prepare thoroughly by studying all course materials.

Practice Midterm

Part A: Multiple Choices

A1 Which of the following is considered a signal-strength related transmission impairment?

a. Noise

b. Distortion

c. Attenuation

d. Crosstalk

A2 Which statement best explains why Manchester encoding provides synchronization?

a. It uses a single voltage level for each bit.

b. It guarantees at least one transition in every bit period.

c. It eliminates noise in transmission.

d. It reduces bandwidth usage.

A3 A line coding scheme uses three voltage levels: $+V$, 0, and $-V$. Bit 1 alternates between $+V$ and $-V$, while bit 0 is represented by 0. What is the main advantage of this scheme?

a. It reduces baseline wandering.

b. It eliminates the need for synchronization.

c. It uses only one voltage level.

d. It guarantees zero bandwidth usage.

A4 What is Address Resolution Protocol?

a. ARP is a data encryption protocol used to secure communication between network devices.

b. ARP is a routing protocol used to determine the optimal path for data transmission across a network.

c. ARP is a network layer protocol that maps an IP address to a MAC address.

d. ARP is a protocol used to compress data packets for efficient transmission over a network.

IP address = logical address
MAC address = physical address

ARP converts IP → MAC

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A5 A sine wave has a frequency of 2 kHz. What is its period?

a. 0.5 ms

b. 2 ms

c. 2000 ms

d. 1 ms

A6 A signal experiences a power loss by a factor of 100. What is the attenuation in dB?

a. -10 dB

b. -20 dB

dB=10log10(P12/P1)

c. -30 dB

d. -40 dB

A7 If a modulation scheme transmits 5 bits per signal element, how many signal levels are used?

r = log2L

a. 16

r= 5
L= 2^5

b. 32

c. 64

d. 128

A8 Which digital-to-analog technique varies the phase of the carrier signal to represent data?

a. ASK

b. FSK

c. PSK

d. AMI

ASK → Amplitude changes

FSK → Frequency changes

PSK → Phase changes

AMI → Line coding (digital-to-digital, not modulation)

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Part B: Short Questions

B1 Explain the main disadvantage of Synchronous TDM in terms of efficiency. Why does this occur?

due to empty time slots.

This occurs because each input channel is assigned a fixed time slot in every frame, even if the channel has no data to send.
As a result, bandwidth is wasted when some sources are idle.

B2 Explain the concept of Protocol Layering and its primary benefit in network design.

Protocol layering is a design approach where network communication is divided into separate layers, and each layer performs a specific function.
Each layer provides services to the layer above and uses services from the layer below.

The primary benefit of protocol layering is modularity, which simplifies design, implementation,
and troubleshooting by separating complex network tasks into manageable components.

B3 Explain the difference between the Time-Domain and Frequency-Domain plots of a signal.

A time-domain plot shows how a signal's amplitude changes with respect to time. It represents the signal as a function of time.

A frequency-domain plot shows how much of each frequency component is present in the signal. It represents amplitude (or power) as a function of frequency.

B4 What is the difference between Baseband Transmission and Broadband Transmission?

Baseband transmission sends digital signals directly over a low-pass channel whose bandwidth starts at 0 Hz and does not use a carrier signal.

Broadband transmission uses modulation to send signals over a bandpass channel centered around a carrier frequency, and the bandwidth does not start at 0 Hz.

B5 What are DC Components in line coding, and why are they considered a problem?

DC components in line coding refer to a non-zero average voltage in the signal, meaning it contains a zero-frequency component.

They are a problem because many transmission media cannot carry DC signals properly, which can cause distortion and signal degradation.

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B6 What is the purpose of the Address Resolution Protocol (ARP)?

is to map an IP address to its corresponding MAC address within a local network.
It allows devices to find the physical address needed to deliver a frame.

Part C: Long Questions

C1 A cable has a signal gain of +0.3 dB/km. If a signal with a power of 2 mW is injected at the beginning, what is the power of the signal after 5 km? Show your steps.

Input power = 2 mW
Distance = 5 km
Gain per km = +0.3 dB

dB = 10 log10(P_2 / P_1)

Total gain = 0.3 × 5 = +1.5 dB

0.15 = log10(P_2 / 2)

P_2 / 2 = 10^0.15 10^0.15 1.41 P_2 = 2 × 1.41
P_2 ≈ 2.82 mW

C2 A signal is digitized using PCM with 128 quantization levels. The original low-pass analog signal has a bandwidth of 200 MHz. Assume Nyquist sampling, $r = 1$ (NRZ line coding), and $c = \frac{1}{2}$. Determine the minimum required bandwidth for the digital signal.

L = 128

bandwidth = 200 MHz

r = 1 (NRZ)

c = 1/2

128= 2^7, so nb =7 bits
fs ≥ >= 2fm
fm = highest frequency in the analog signal.....fs = 2 × 200 MHz.

N = nb × fs
N = 7 × 400 MHz
N = 2800 Mbps
N = 2.8 × 10^9 bps

B = c × N × (1/r)
B = (1/2) × 2.8 × 10^9
B = 1.4 × 10^9 Hz

Final Answer:

The minimum required bandwidth for the digital signal is

Bmin = 1.4 GHz

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C3 A synchronous TDM multiplexer combines 4 input channels. Each input channel has a data rate of 1 Mbps. The input unit is 1 bit. Each frame also contains 1 synchronizing bit.

a. Determine the frame size (in bits).

b. Determine the frame rate.

c. Determine the output bit rate.

d. Determine the frame duration.

e. Determine the bit duration of the output link.

a. Frame size = 4 + 1
Frame size = 5 bits

b. Frame rate = 1,000,000 frames per second

c. Output bit rate = 5 × 1,000,000
Output bit rate = 5,000,000 bits/s

Tframe =1 /frame rate=1 /1,000,000=10^−6 s= 1 microsecond (1 μs)

d. Frame duration = 1 / frame rate
Frame duration = 1 / 1,000,000

Frame duration = 1 microsecond (1 μs)

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